

Tide retrospective: the weighted-jet recovery theorem (weighted-jet programme, stages 3D–3G)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The programme’s closing tide. The pairwise difference limit of the previous stage reduces the germbij §7.4 recovery to bookkeeping: a quotient rule, a variance-positivity bridge, a strong induction, and a change of variables. This tide executes those four stages — and, in the process, discovered and repaired a serious process failure in the previous tide.

2 The incident

At this tide’s first build, the previous tide’s file failed with fifteen errors — a file whose tide log and retrospective claimed a clean build with zero warnings. The claim was false: that tide never added its file to `Laplace.lean`’s imports, so its `lake build` succeeded vacuously (the file was outside the import closure), local audits saw nothing, and PR #55’s CI passed for the same reason, merging an uncompiled file to `main`. The errors were all ordinary first-build classes (renames, beta-unreduced congruence goals, implicit-argument inference, dead tactics) and were repaired here; the mathematics was unaffected. The prior retrospective now carries an erratum, the tide log records the incident, and the repository `CLAUDE.md` carries the new rule: *a build result may be reported only after verifying both the import line and the new .olean*. The deeper lesson is that the build gate silently weakens to a no-op exactly when a step is forgotten, so the gate itself must be checked, not just its exit code.

3 The thing formalised

Six declarations in `Laplace/OneD/JetRecovery.lean`, sorry-free with zero warnings (verified via import and `.olean` this time). The chain: the base-point limit $J_s(q) \rightarrow A_s$; the generic quotient-difference lemma (limits of scaled numerator and denominator differences give the scaled limit of the quotient difference, over any filter, with Lean’s division-by-zero convention handling the scale degenerately); the variance bridge $A_{2m}A_0 - A_m^2 > 0$ from stage 1 at $t = a(2k)!$; the one-rung recovery; the strong induction; and the headline¹:

For two finite jets $ax^{2k} + \sum_{r=1}^R c_r^\nu x^{2k+r}$ with uniformly positive profiles, if the Gibbs expectations of x^{2k+r} , $1 \leq r \leq R$, eventually agree as $t \rightarrow \infty$, then $c^1 = c^2$.

¹`polynomialJet.recovery`.

With the pure-monomial recovery of (k, a) from the earlier recovery thread, this completes the constructive dimension-one programme of germbij §7.4(a) for finite jets: every coefficient of the potential is pinned by moment asymptotics, each at its own order, with the injectivity supplied by the variance of the matching observable.

4 Proof notes

At rung r with $m = 2k + r$, the quotient-difference lemma applied to J_m and Z produces the limit $-\delta_r(A_{2m}A_0 - A_m^2)/A_0^2$; uniqueness of limits against the data hypothesis and the variance bridge force $\delta_r = 0$. The transfer to t -data composes the stage-2 exact scaling identity with the observation that $q \mapsto (q^{2k})^{-1}$ carries $\mathcal{N}[>]0$ to **atTop**, so eventual equality of t -moments becomes eventual vanishing of the scaled q -difference. Three tactical patterns recurred often enough to promote: **set** folds only occurrences existing at the time (hypotheses created later arrive unfolded — refold with the definitional equation); **Tendsto.congr** produces beta-unreduced lambda applications (prove the function equality by **funext** and **rwa** instead); and **div_eq_zero_iff** with **resolve_right** cleanly clears a positive denominator from an equation to zero.

5 The deliberation

No fresh consult: stages 3D–3G were specified with proof sketches in the route consult, whose prediction (“once the unnormalized limit is available, the covariance quotient and strong induction should be comparatively routine”) held — the mathematics went through essentially as sketched, with the incident being process rather than mathematics. The one addition was the base-point limit, implicit in the consult’s quotient rule.

6 Mathlib vs new

Mathlib lookup: the filter dominated convergence theorem again, eventual nonvanishing along a limit, uniqueness of limits, the inverse map carrying $\mathcal{N}[>]0$ to **atTop**, and strong induction on naturals.² Seabed reuse: everything from stages 1–3C. New: the assembly.

7 What the next tide inherits

The weighted-jet programme is complete: six tides from scoping to the recovery theorem, against the consult’s original eight-stage plan, with the middle replaced by the pairwise-difference route at half the machinery. Natural next steps for the germbij commissions: the **leanref** markers for the programme on the note’s §7.4(a) (after this PR merges), and then either the general-order nondegenerate induction (now de-risked by this programme’s normalized-series handling) or the multivariate first rung. Both need fresh scoping.

²Respectively **Tendsto.eventually_ne**, **tendsto_nhds_unique**, **Filter.Tendsto.inv_tendsto_nhdsGT_zero**, and **Nat.strong_induction_on**.